

Robin Connor Schramm

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Research Interests

My overarching goal is to enable meaningful interaction, visualization, and information dissemination through novel multimodal interfaces and technologies. My multidisciplinary research in Human-Computer Interaction (HCI) is deeply rooted in experimental methods and the design, implementation, and evaluation of interactive systems. A significant portion of my work involves building functional prototypes for realistic use contexts and studying them empirically to understand how people interact with emerging technologies beyond controlled lab conditions. My current work focuses on creating and evaluating novel interfaces that can support future forms of human-AI interaction.

Education

RheinMain University & Mercedes-Benz Tech Innovation *Dec. 2022 – present*

PhD in Human-Computer Interaction

Advisor: [Prof. Dr. Ulrich Schwanecke](#)

Thesis Title: Interactive Points of Interest for In-Car Augmented Reality

University College London (UCL) *Mar. 2026 – May 2026*

Visiting Researcher

Host: [Dr. Mark Colley](#)

Supported by the German Academic Exchange Service (DAAD) [HAW.International](#) program.

Reutlingen University *Sep. 2019 – Sep. 2021*

Master of Science in Human-Centered Computing

Thesis Title: Simulation of Long-Term SLAM for Mobile Robots

Thesis Cooperation: [Fraunhofer IPA](#)

Reutlingen University *Mar. 2016 – Aug. 2019*

Bachelor of Science in Computer Science and Media

Thesis Title: Comparison of Motion Tracking and Inverse Kinematics for Presence in Virtual Reality

Thesis Cooperation: [Fraunhofer IAO](#)

Job Experience

Human-Computer Interaction Group, University of Konstanz *Jun. 2026 – present*

Postdoctoral Researcher

Constance, Germany

Research on Human-AI interaction in mixed reality.

Mercedes-Benz Tech Innovation, XReality *Oct. 2021 – May 2026*

Software Engineer

Böblingen, Germany

I developed prototypes and novel interactive systems in fast-changing environments with evolving hardware requirements, specializing in in-car Augmented Reality applications using Unity. My work also included testing in labs, in the field, and through empirical studies. Additionally, I created showcases and demos to demonstrate the capabilities of these systems.

Fraunhofer IPA *Feb. 2021 – Sep. 2021*

Research Assistant

Stuttgart, Germany

I developed a virtual twin-like simulation for visual SLAM for mobile robots using Unity. This involved creating realistic simulations to test and refine visual SLAM algorithms, ensuring they perform effectively in dynamic and changing environments.

Reutlingen Research Institute *Mar. 2020 – Feb. 2021*

Research Assistant

Reutlingen, Germany

In the [Project KI Delta Learning](#), I developed Unity simulations using photogrammetry data and applied com-

puter vision techniques to generate synthetic data for autonomous driving.

Fraunhofer IAO
Research Assistant

Aug. 2019 – Feb. 2020
Stuttgart, Germany

I developed innovative tools for collaborative Virtual Reality applications, specifically tailored for construction data visualization and analysis.

Bizerba
Working Student

Feb. 2018 – Jan. 2019
Balingen, Germany

I developed user interfaces for embedded systems and created new Linux build pipelines for long-term support systems.

Teaching

Lecturer - Scientific Writing
Reutlingen University

Oct. 2023 – Jul. 2026
Reutlingen, Germany

As a Lecturer for the seminar "Topics in Computer Science" I guide students through the basics of the scientific process, citation practices, systematic literature reviews, and academic writing. I provide detailed feedback on their papers and teach the essentials of delivering research talks and using AI tools. I have reviewed and graded papers and talks of over 150 students over the course of six semesters.

Tutor - Basics in Programming
Reutlingen University

Mar. 2019 – Feb. 2021
Reutlingen, Germany

Led the Programming Fundamentals Lab, where I taught students the basics of programming in C and Python through interactive live coding sessions, comprehensive reviews, and engaging Q&A discussions. Additionally, I reviewed exam questions and assisted in grading coding exams.

Selected Publications

1. **Robin Connor Schramm**, Ginevra Fedrizzi, Markus Sasalovici, Jann Philipp Freiwald, and Ulrich Schwanecke. 2025. *Augmented Journeys Interactive Points of Interest for In-Car Augmented Reality*. In CHI Conference on Human Factors in Computing Systems (CHI '25), ACM, doi: [10.1145/3706598.3714323](https://doi.org/10.1145/3706598.3714323)
🏆 Received a **CHI Honorable Mention Award for Best Paper** (top 5%)
2. **Robin Connor Schramm**, Markus Sasalovici, Jann Philipp Freiwald, Michael Otto, Melissa Reinelt, and Ulrich Schwanecke. 2025. *Blending the Worlds World-Fixed Visual Appearances in Automotive Augmented Reality*. In CHI Conference on Human Factors in Computing Systems (CHI '25), ACM, doi: [10.1145/3706598.3713185](https://doi.org/10.1145/3706598.3713185)
3. Markus Sasalovici, Albin Zeqiri, **Robin Connor Schramm**, Oscar Javier Ariza Nunez, Pascal Jansen, Jann Philipp Freiwald, Mark Colley, Christian Winkler, and Enrico Rukzio. 2025. *Bumpy Ride? Understanding the Effects of External Forces on Spatial Interactions in Moving Vehicles*. In CHI Conference on Human Factors in Computing Systems (CHI '25), ACM, doi: [10.1145/3706598.3714077](https://doi.org/10.1145/3706598.3714077)
4. **Robin Connor Schramm**, Markus Sasalovici, Axel Hildebrand, and Ulrich Schwanecke. 2023. *Assessing Augmented Reality Selection Techniques for Passengers in Moving Vehicles A Real-World User Study*. In ACM International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI '23), ACM, doi: [10.1145/3580585.3607152](https://doi.org/10.1145/3580585.3607152)
5. Markus Sasalovici, Stephan Leenders, **Robin Connor Schramm**, Jann Philipp Freiwald, Hannes Frederic Botzet, Daniel Keßelheim, Thomas Krach, and Christian Winkler. 2023. *In-Car Office: Can HMD-Based AR Alleviate Passenger Motion Sickness?*. In 15th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI '23 Adjunct), ACM, doi: [10.1145/3581961.3609869](https://doi.org/10.1145/3581961.3609869)

Service and Volunteering

- Associate Chair for the [NordiCHI'26](#) Late-Breaking Work track
- Associate Chair for the [MuC'26](#) Full Papers track
- Part of the Organizing Committee as Proceedings Chair for [MUM'26](#)

- I regularly review papers at top HCI venues such as CHI, MobileHCI, AutomotiveUI, ACM ISS, and the International Journal of Human-Computer Studies
- Student Volunteer at IEEEVR '21

Thesis Supervision

Master Theses as main supervisor (all at Mercedes-Benz Tech Innovation):

- Leo Kruse, University of Porto (2025)
- Alireza Parchami, Saarland University (2025)
- Hongcheng Jia, University of Stuttgart (2024)
- Ginevra Fedrizzi, University of Trento (2024)

Talks, Seminars, and Demos

- Ulm University (2025): "In-Car Mixed Reality - current research and challenges" - invited talk at the Automotive User Interfaces Lecture
- Mercedes-Benz Tech Innovation (2025): "Success Story - Research in Automotive User Interfaces" - in-person presentation at the company townhall meeting
- Mercedes-Benz PhDs poster session [Doktoranden Marktplatz](#) (2024) - in-person
- WeAreDevelopers World Congress - [Mercedes-Benz booth](#) (2023): "In-Car Augmented Reality with the Varjo XR-3" - in-person interactive live demonstration
- Reutlingen University (2023): "Automotive User Interfaces in HCI" - in-person
- Mercedes-Benz AG (2023): "Augmented Reality Selection Techniques in Moving Vehicles" - digital research cluster talk
- Mercedes-Benz Tech Motion (2022): "Integration of Head-Mounted Displays (HMDs) in Cars" - in-person tech talk

Skills

- **Coding:** Mainly *C#* for prototyping and creating interactive XR applications and user-studies, especially with Unity for both Windows and Android; *R* for statistical analysis and data visualization; also proficient in *C*, *Python*, and *Java*
- **AI-Assisted Development:** Proficient in leveraging various AI coding tools in a fast paced environment for code generation, refactoring, debugging, and technical documentation, with rigorous validation of outputs for quality and reliability
- **Industry Projects:** Proficient in working in teams with agile frameworks like Scrum and managing tasks with Kanban boards and Jira; regularly use and maintain *GitHub* repositories
- **Research:** Planning, building, and conducting *user studies*, both in the lab and in the field; *Systematic Literature Review* for comprehensive analysis and synthesis of existing research; analyzing and interpreting both *qualitative* and *quantitative data*; *Writing and Reviewing* research papers in HCI and Computer Science
- **Languages:** *German* – Native; *English* – Proficient